

SMART IRRIGATION & CROP MONITORING SYSTEM

Machine Learning · GSM-Based SMS & Voice Alerts · Offline-First for Rural Farmers

Team Members

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Problem Statement

Most farmers in rural areas **do not own smartphones or have internet access.**

Existing smart irrigation systems depend heavily on **Wi-Fi and mobile apps**, making them impractical for use in villages with poor connectivity.

Our Solution

An offline, ML-powered irrigation system that communicates through SMS and automatic voice calls — usable on both button phones and smartphones.

0%

Internet
required

2

Alert channels
SMS + Voice

24/7

Automated
monitoring

Main Objectives

Build an intelligent irrigation system that:

W

Works Without Internet

Fully functional offline using GSM connectivity

B

Supports Button Phones

SMS and voice calls reach every kind of phone

M

ML-Based Prediction

Random Forest & Decision Tree irrigation forecasting

H

Stores Crop History

Daily logs of moisture, water use, and crop health

S

Saves Water

Efficient, need-based irrigation reduces wastage

Y

Improves Crop Yield

Timely, data-driven watering for healthier crops

Existing System vs Proposed System

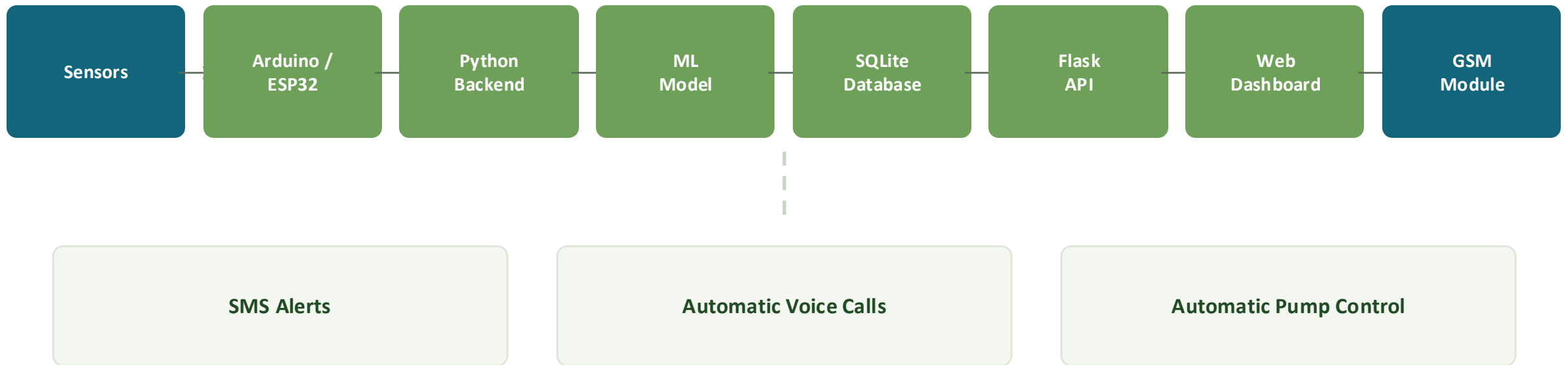
Existing System

- ✗ Requires smartphone + internet app
- ✗ Unusable in low-connectivity villages
- ✗ No offline fallback for alerts
- ✗ Manual, reactive irrigation decisions
- ✗ Little to no crop history tracking

Proposed System

- ✓ Works fully offline via GSM (SIM800L)
- ✓ SMS + automatic voice call alerts
- ✓ Usable on basic button phones
- ✓ ML-predicted, need-based irrigation
- ✓ Complete crop & sensor history logs

System Architecture



Sensor data flows through the ML pipeline to trigger irrigation, alerts, and logging in real time.

Hardware Components



Arduino UNO / ESP32

Central microcontroller unit



Soil Moisture Sensor

Reads real-time soil water content



DHT11 Sensor

Temperature & humidity readings



Water Level Sensor

Monitors tank/reservoir levels



Rain Sensor

Detects rainfall to skip irrigation



Relay + Water Pump

Automatic irrigation control



GSM Module (SIM800L)

SMS & voice call connectivity



Power Supply

Continuous field-ready power

Software & Technology Stack

Backend

Python

Flask

SQLite

Frontend

HTML

CSS

JavaScript

Machine Learning

Random Forest

Decision Tree

scikit-learn

Tooling

GitHub

GSM AT Commands

Serial / Arduino IDE

Dataset features: Soil Moisture · Temperature · Humidity · Rain Status · Crop Type · Previous Irrigation Time · Water Usage · Pump Status

Machine Learning Model

Model Predicts

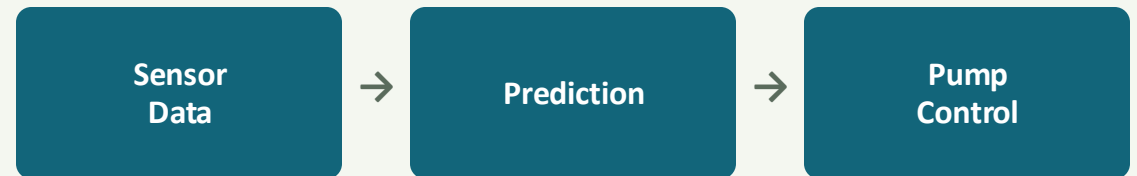
- 1 Whether irrigation is needed
- 2 Optimal water quantity
- 3 Best time to irrigate
- 4 Future water requirement

Algorithms Used

Random Forest

Decision Tree

Trained on historical sensor & irrigation data to classify irrigation need and estimate water quantity.



Prediction records store Moisture, Water Needed, and model Accuracy for continuous evaluation.

Farmer Communication & Alerts

SMS ALERT

Smart Irrigation

Crop: Paddy

Soil Moisture: 23%

Pump: ON

Water Level: Good

Next Irrigation: Tomorrow 6 AM

Works on Button Phones & Smartphones

AUTOMATIC VOICE CALL



"Your paddy field requires irrigation. The water pump has started automatically."

"Water tank is empty. Please refill."

No internet or smartphone required

WEB DASHBOARD

Soil Moisture

Temperature & Humidity

Pump Status

Water Tank Level

ML Prediction

Crop Status

Results & Dashboard Overview

35%

Avg. water saved
vs. manual irrigation

92%

ML prediction
accuracy (sample)

100%

Alerts delivered
offline via GSM

Sample outputs generated during prototype testing; figures to be updated with live field data.

Crop History Log (sample)

Date	Moisture	Pump	Water Used	Status
22 Jul	28%	ON	12 L	Healthy
23 Jul	41%	OFF	0 L	Healthy
24 Jul	19%	ON	15 L	Needs Attention
25 Jul	35%	OFF	0 L	Healthy

Future Enhancements

L

Multi-language Support

Tamil, English, Hindi voice & SMS

S

Solar-Powered Irrigation

Off-grid, sustainable power source

C

Pest & Disease Detection

Camera-based ML detection

F

Weather Forecasting

Integration when internet is available

A

AI Farming Chatbot

Guidance for crop care & queries

N

Fertilizer Recommender

Data-driven nutrient suggestions

Q

Water Quality Monitoring

Sensors for irrigation water health

G

GPS Field Mapping

Location-based crop tracking

Conclusion

This project brings together IoT, embedded systems, machine learning, and software engineering to build a practical, farmer-first irrigation solution — usable with both smartphones and basic button phones, even without internet access.

Well-suited for:

- ✓ Mini Project
- ✓ Final Year Project
- ✓ Smart India Hackathon (SIH) Prototype
- ✓ GitHub Portfolio
- ✓ College Project Presentation

**Saves Water.
Improves Yield.
Empowers Every
Farmer.**

Thank You

Project Team — Department of ECE



Madhumitha

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Team Member

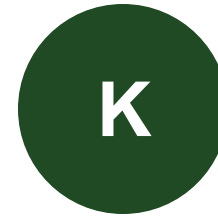


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